

## Algebra 1 Honors Unit 3 - Part 2 Assessment Guide

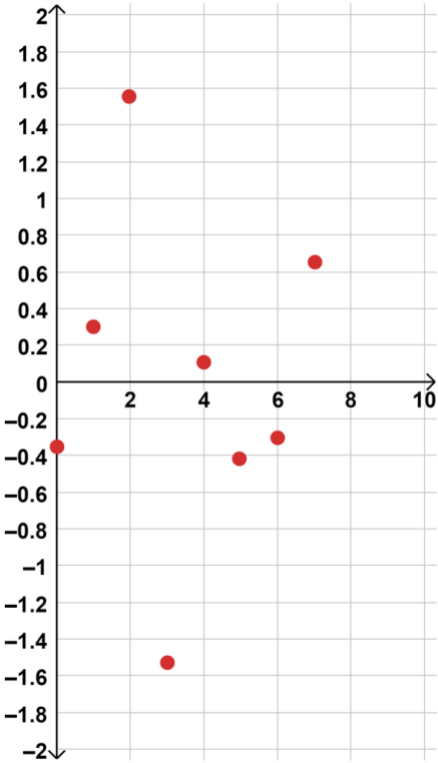
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| <p style="text-align: center;"><b>General Information</b></p>      | <p>The Unit 3 assessment is divided into two parts. Part 1 assesses benchmarks MA.912.AR.1.2 and MA.912.AR.2.2, and Part 2 assesses MA.912.DP.2.4, MA.912.DP.2.6, and the honors benchmark MA.912.DP.2.5. Teachers familiar with the MAFS standards may not immediately notice the distinction between MA.912.DP.2.6, and the honors benchmark MA.912.DP.2.5. MA.912.DP.2.6 states “given a scatter plot with a line of fit and residuals” whereas MA.912.DP.2.5 states “given a scatter plot that represents bivariate numerical data”. The subtle difference in what a student is given is important for teachers to recognize. Honors Algebra 1 teachers should note that students are required to plot the residual values.</p> <p>The benchmarks MA.912.DP.2.4 and MA.912.DP.2.6 build on students’ understanding of scatter plots from Grade 8 or Grade 7 Accelerated where students learned to construct scatter plots (MA.8.DP.1.1), describe patterns of association (MA.8.DP.1.2), and informally fit a straight line to a scatter plot with a linear association (MA.8.DP.1.3).</p> <p>Students’ work in Unit 3 with scatter plots focuses on interpreting the slope and <math>y</math>-intercept of linear models fit to data and using those models to solve real-world problems as well as determining and interpreting the strength and direction of the correlation in real-world contexts. Instruction in this unit also includes formally fitting a linear function to data using technology. While this is included in instruction, it is not included on this assessment as students will not have access to such technology on state assessments. For honors students, they also explore how to assess the fit of a linear function by plotting and analyzing residuals. Question 4 assesses this benchmark.</p> <p>Students will build on this work in Unit 4 when working with linear functions in a variety of real-world context. Data from this assessment can be used to determine potential supports students may need when interpreting linear functions in context. For students needing remediation support for these benchmarks, consider using resources from Grade 8, Unit 11 to scaffold.</p> |
| <p style="text-align: center;"><b>Calculator Information</b></p>   | <p>The assessment is calculator neutral.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p style="text-align: center;"><b>Total Recommended Points</b></p> | <p>The total recommended points in this guide is 99 points. Teachers can either determine where best to use the extra point or teachers may round the total value up to 100 points.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Question                                                                                                               | 1a       | Benchmark(s)                            | MA.912.DP.2.4                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| The slope represents the change in total weight of the bowl and liquid, in grams, for each tablespoon of liquid added. |          | Recommended Scoring<br>(8 Total Points) | Consider no partial credit.                                                                                             |
| Question                                                                                                               | 1b       | Benchmark(s)                            | MA.912.DP.2.4                                                                                                           |
| The y-intercept represents the total weight of the bowl, in grams, before any liquid is added.                         |          | Recommended Scoring<br>(8 Total Points) | Consider no partial credit.                                                                                             |
| Question                                                                                                               | 1c       | Benchmark(s)                            | MA.912.DP.2.4                                                                                                           |
| Weight (g)                                                                                                             | 1127.32  | Recommended Scoring<br>(8 Total Points) | Consider no partial credit.                                                                                             |
| Question                                                                                                               | 1d       | Benchmark(s)                            | MA.912.DP.2.4                                                                                                           |
| Amount of Liquid (tablespoons)                                                                                         | about 23 | Recommended Scoring<br>(8 Total Points) | Consider no partial credit. Students may approximate differently, but should receive credit as long as it's reasonable. |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2                          | Benchmark(s)         | MA.912.DP.2.4              |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------|----------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <table border="1"><caption>Data points from the scatter plot</caption><thead><tr><th>Years since 2004 (x)</th><th>Sales (millions of \$) (y)</th></tr></thead><tbody><tr><td>0.5</td><td>180</td></tr><tr><td>1.5</td><td>200</td></tr><tr><td>2.5</td><td>240</td></tr><tr><td>3.5</td><td>260</td></tr><tr><td>4.5</td><td>250</td></tr><tr><td>5.5</td><td>310</td></tr><tr><td>6.5</td><td>360</td></tr><tr><td>7.5</td><td>370</td></tr></tbody></table> |                            | Years since 2004 (x) | Sales (millions of \$) (y) | 0.5                                                    | 180                                                                                                                                                                 | 1.5 | 200 | 2.5 | 240 | 3.5 | 260 | 4.5 | 250 | 5.5 | 310 | 6.5 | 360 | 7.5 | 370 | <b>Recommended Scoring</b><br><b>(5 Total Points)</b> | Consider no partial credit. Any trend line that follows the trend of the data should be considered correct. |
| Years since 2004 (x)                                                                                                                                                                                                                                                                                                                                                                                                                                          | Sales (millions of \$) (y) |                      |                            |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
| 0.5                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 180                        |                      |                            |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
| 1.5                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 200                        |                      |                            |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
| 2.5                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 240                        |                      |                            |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
| 3.5                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 260                        |                      |                            |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
| 4.5                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 250                        |                      |                            |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
| 5.5                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 310                        |                      |                            |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
| 6.5                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 360                        |                      |                            |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
| 7.5                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 370                        |                      |                            |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2b                         | Benchmark(s)         | MA.912.DP.2.4              |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
| <table border="1"><tr><th>Equation</th><td><math display="block">y = \frac{100}{3}x + 160</math></td></tr></table>                                                                                                                                                                                                                                                                                                                                            |                            | Equation             | $y = \frac{100}{3}x + 160$ | <b>Recommended Scoring</b><br><b>(10 Total Points)</b> | Consider where the student drew the trend line for question 2a. Consider partial credit for students who have either the slope or y-intercept correct but not both. |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |
| Equation                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $y = \frac{100}{3}x + 160$ |                      |                            |                                                        |                                                                                                                                                                     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                       |                                                                                                             |

| Question                                                                                                                                                                                                                                                                                                                                                                  | 2c | Benchmark(s)                                           | MA.912.DP.2.4                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| The slope is the change in total sales, in millions of dollars, for each year since 2004.                                                                                                                                                                                                                                                                                 |    | <b>Recommended Scoring</b><br><b>(8 Total Points)</b>  | Consider no partial credit.                                                                                       |
| Question                                                                                                                                                                                                                                                                                                                                                                  | 2d | Benchmark(s)                                           | MA.912.DP.2.4                                                                                                     |
| The $y$ -intercept represents the total sales, in millions of dollars, for the year 2004.                                                                                                                                                                                                                                                                                 |    | <b>Recommended Scoring</b><br><b>(8 Total Points)</b>  | Consider no partial credit.                                                                                       |
| Question                                                                                                                                                                                                                                                                                                                                                                  | 3a | Benchmark(s)                                           | MA.912.DP.2.6                                                                                                     |
| <p>The data for The Netherlands has a</p> <div> <input type="radio"/> positive<br/> <input type="radio"/> negative </div> <p>correlation because</p> <div> <input type="radio"/> the slope of the line of best fit is<br/> <input type="radio"/> most of the residual values are </div> <div> <input type="radio"/> positive.<br/> <input type="radio"/> negative. </div> |    | <b>Recommended Scoring</b><br><b>(10 Total Points)</b> | Students should complete all pieces of the statement correctly to show understanding. Consider no partial credit. |

| Question | 3b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Benchmark(s)                                                      | MA.912.DP.2.6                                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <p> <input type="radio"/> Germany<br/> <input type="radio"/> The Netherlands         </p> <p>has a</p> <p> <input type="radio"/> weaker<br/> <input type="radio"/> stronger         </p> <p>correlation than</p> <p> <input type="radio"/> Germany<br/> <input type="radio"/> The Netherlands         </p> <p>because the residuals have a</p> <p> <input type="radio"/> larger<br/> <input type="radio"/> smaller         </p> <p>absolute value, indicating that the</p> <p>distance each</p> <p> <input type="radio"/> actual<br/> <input type="radio"/> predicted         </p> <p>value is from the line of best fit is</p> <p> <input type="radio"/> larger.<br/> <input type="radio"/> smaller.         </p> <p style="text-align: center; color: red; font-weight: bold;">OR</p> <p> <input type="radio"/> Germany<br/> <input type="radio"/> The Netherlands         </p> <p>has a</p> <p> <input type="radio"/> weaker<br/> <input type="radio"/> stronger         </p> <p>correlation than</p> <p> <input type="radio"/> Germany<br/> <input type="radio"/> The Netherlands         </p> <p>because the residuals have a</p> <p> <input type="radio"/> larger<br/> <input type="radio"/> smaller         </p> <p>absolute value, indicating that the</p> <p>distance each</p> <p> <input type="radio"/> actual<br/> <input type="radio"/> predicted         </p> <p>value is from the line of best fit is</p> <p> <input type="radio"/> larger.<br/> <input type="radio"/> smaller.         </p> | <p><b>Recommended Scoring</b></p> <p><b>(10 Total Points)</b></p> | <p>There are two possible correct answers. Students should complete all pieces of the statement correctly to show understanding. Consider no partial credit.</p> |

| Question                                                                                                                                                                  | 4a                                                                                | Benchmark(s)                                                  | MA.912.DP.2.5                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                           |  | <p><b>Recommended Scoring</b><br/><b>(8 Total Points)</b></p> | <p>Consider 1 point for each plotted point.</p>                                                                                                                                  |
| Question                                                                                                                                                                  | 4b                                                                                | Benchmark(s)                                                  | MA.912.DP.2.5                                                                                                                                                                    |
| <p>The function is a good fit for the data set because the residuals are randomly scattered and there is a balance of points above and below the <math>x</math>-axis.</p> |                                                                                   | <p><b>Recommended Scoring</b><br/><b>(8 Total Points)</b></p> | <p>Consider 4 points for stating that the residuals are randomly scattered and 4 points for stating there is a balance of residuals above and below the <math>x</math>-axis.</p> |